

HOW DOES INTEREST RATE AFFECT HOUSING PRICES?

DATA

- ❖ Federal Funds Rates (from Federal Reserve Economic Data (FRED))
- ❖ Housing Prices (from Zillow)
- ❖ Time Period: 2021 - 2026 (monthly data)

MODEL

- ❖ $\text{Price} = \beta_0 + \beta_1 \cdot \text{interest_rate} + \varepsilon$

$$\text{Price} = 199,100 + 3069 \cdot \text{interest_rate} + \varepsilon$$



KEY FINDINGS:

- ❖ There is a positive relationship between interest rate and housing prices.
- ❖ A 1% increase in interest rates, increases the housing prices by about \$3,069.
- ❖ This pattern is likely due to both changing over time

OLS ASSUMPTIONS:

- ❖ Residual plot shows a clear pattern
- ❖ Linearity: Not satisfied, curved relationship
- ❖ Independence: Not satisfied, residuals are related over time (time series data)
- ❖ Homoscedasticity: Not satisfied, residuals have uneven spread
- ❖ Normality: Not satisfied, residuals are not bell-shaped (skewed distribution)

CONCLUSION:

- ❖ Interest rate and housing prices seem to positively correlate to each other but this doesn't mean one causes the other.
- ❖ Housing prices can be also affected by other factors like the economy.
- ❖ So, we should be careful in the way we interpret these results.